The Altermagnetism in RuO_2

by

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Section: Condensed Matter and Statistical Physics
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Abstract

In this work, we investigate and confirm the existence of altermagnetism as a third state of magnetism in RuO_2. We also present the procedure for calculating the Berry curvature, illustrating the trace of the Dirac point in the RuO_2 band structure. Testing the stability of the system under doping, we found and had a discussion on the reduction of the on-site magnetisation and Fermi level in electron doping, and its stability under hole doping.

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Contents

1 Introduction

Traditionally, the magnetic materials' states are divided into two categories based on their spin structure and properties: ferromagnetic and antiferromagnetic materials. Ferromagnetic materials have a large net magnetisation, which has been known and used by humankind for ages. The favour of one spin over the other is reflected by the spin-polarisation in the electronic structure, with one spin having a lower energy than the other, therefore favouring it. This polarised band structure also indicates the time reversal symmetry breaking of the system, which in turn is taken into account in many other conductivity properties of the materials. The antiferromagnetic, on the other hand, has been found more recently to have a small vanishing net magnetisation. This phenomenon occurs due to individual magnetic moments cancelling each other out by the anti-parallel spin structure. As a result, this structure can be hard to distinguish from the non-magnetic material when using macroscopic electronic and optical probes for magnetic measurements.

Recently, a third magnetic state has been discovered and confirmed, called "Altermagnetism". Experimental results showing that altermagnetism materials exhibit the spin-polarised band structure but an anti-aligned structure of spin.

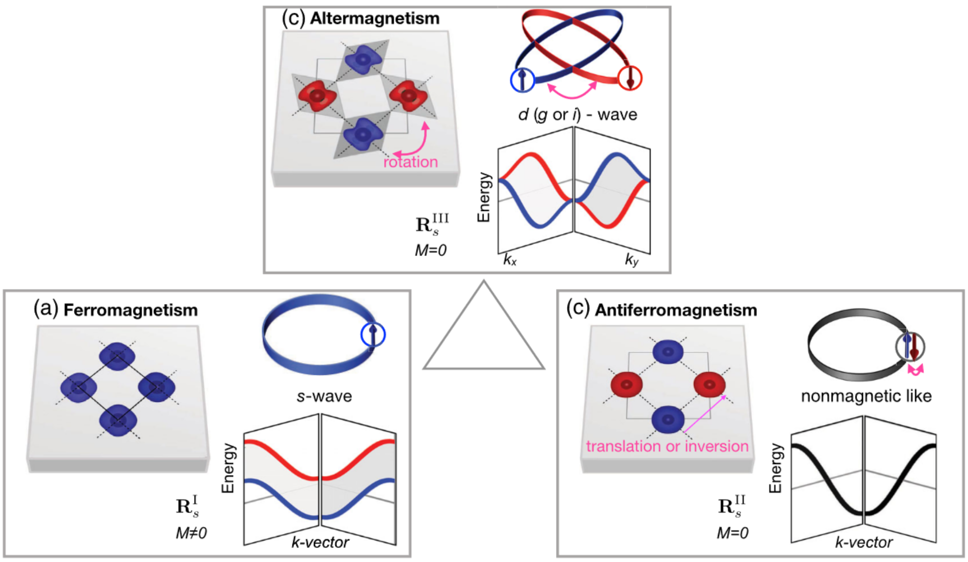


Figure 1: Comparison between the three states of magnetic material in the non-relativistic case

In the comparison shown in the Fig. 1, the ferromagnetic (1a) have a one conventional lattice with an isotropic Fermi surface and the time-symmetry breaking spin-polarisation in the electronic structure. In contrast, the conventional antiferromagnetic (1b[1]) consists of two interlinked sub-lattices with opposite spins, which are connected through translation or inversion and then followed by a spin flip to recover the same as before. This arrangement results in no polarisation in k-space and zero net magnetisation in real space. The altermagnetic material 1c, on the other hand," (this is a joke :v), possesses the unique features of both ferromagnetic and antiferromagnetic states, displaying an anti-parallel structure in real space, but connected through rotation between the two sub-lattices instead of through translation or inversion. Their electronic structure also indicates a breaking of time-reversal symmetry as the ferromagnetic material, which is evident in the spin-splitting band structure.

Recent theoretical works have pointed out that the anisotropies of the Fermi surfaces in the altermagnetism is in fact related to the d-wave, not original well known electron orbital binding to the atom, but raised the question and being answered to be the long-sought counterpart of the unconventional d-wave superconductor[1], indicate the deep connection between the superconductivity and magnetism. Various

[1] This typing error comes from the paper, not my

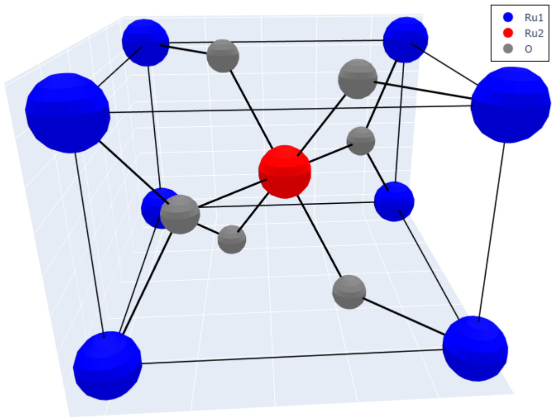


Figure 2: Structure of RuO_2, Ru atom colored according to their magnetic moment, showing the antiferromagnetic structure.

theories have been proposed to resolve the contradictions observed between antiferromagnetic structure and ferromagnetic phenomena, leading to a widely accepted belief that d-wave and even higher-order magnetic waves could provide sufficient explanations. What stands out from these discoveries is the altermagnetic field that emerges from the basic level of single-particle non-relativistic description of collinear magnets, which in turn eventually leads us to the field of the robust, elementary, magnetic phase. Ongoing research aims to deepen the understanding of altermagnetism, exploring its links to many-body correlations, relativistic effects, noncollinear magnetic ordering, topological phenomena, and frustrated magnetic interactions.

While the insights presented here reflect a theoretical perspective on altermagnetism, its experimental confirmation has positioned this field as a promising area for technological applications. One of the most focused candidates in that research is RuO_2, which has been in favour of the physicist in recent years(need cite). Spintronics, which has historically utilised ferromagnetic materials, relies on the separation of spin-polarised bands into majority and minority channels, leading to different conductance levels for these spin currents. This combination of these properties results in the potential of the application of altermagnetism to be the core material in future spintronic devices.

Why should we care about altermagnetism when other materials also exhibit band-splitting? The key lies in quantum mechanics. When we factor in spin-orbit coupling (SOC) in our calculations, spin is no longer a well-defined quantum number; instead, we work with the total angular momentum $\vec{J}$. For example, the spin splitting observed in transition metal dichalcogenides (TMDs) results from SOC perturbation. In contrast, the spin-splitting in altermagnetic materials is based primarily on the symmetry of the lattice rather than relying on SOC, which can diminish spin stability instead of enhancing it.

2 Theory Background

2.1 Particle in Periodic System

A crystallite system can be described using a basis and a Bravais lattice. The lattice point of the system is every point satisfying $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3 \,\forall\, n \in \mathrm{N}$, with the basis $\{\mathbf{a}\}$ of Bravais lattice discussed above, showing the atom inside the lattice unit cell and its relative position in the unit cell.

As we already discussed, the unit cell is the replica of the Bravais lattice all over space, describing the periodicity of the system. Alongside that, it is also worth mentioning the supercell, which is constructed by the replication of the unit cell along one or some of the basis vectors.

The periodic system results in the periodicity of its potential:

$$U(\mathbf{r}) = U(\mathbf{r} + \mathbf{R}),\ \forall\,\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3. \tag{1}$$

A system with translational symmetry in position space $\{\mathbf{r}\}$ has its Fourier transformation as its dual, which is also called k-space. The reciprocal lattice is the sublattice of the k-space, which contains all the vectors $\mathbf{G} = m_1\mathbf{b}_1 + m_2\mathbf{b}_2 + m_3\mathbf{b}_3$, so that:

$$\mathbf{a}_i\mathbf{b}_j = 2\pi\delta_{ij}$$

With this, the reciprocal vectors $\mathbf{G}$ correspond to the periodic of the system:

$$\exp(\mathbf{G}\cdot(\mathbf{r} + \mathbf{R})) = \exp(\mathbf{G}\cdot\mathbf{r}) \tag{2}$$

If the system has the periodic potential (1), then according to the Bloch theorem, the electron wave function can be factorised as:

$$\psi_{\mathbf{k}}(\mathbf{r}) = u_k(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{3}$$

In which, $u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{\mathbf{k}}(\mathbf{r})$, is called *the periodic Bloch function.* As a consequence of the periodicity of the k-space, the Bloch wavefunction $\psi_{\mathbf{k}}(\mathbf{r})$ is only unique within the primitive cell of k-space, called the Brillouin zone (BZ). So, in the calculation of the solid state, we only need to consider points inside the BZ.

2.2 Kohn-Sham Density Functional Theory

Let us start with the Hamiltonian of the full system, which consists of an electron and an ion:

$$H = -\sum_I \frac{\hbar^2\nabla^2_{\mathbf{R}}}{2M_I} - \sum_i \frac{\hbar^2\nabla^2_{\mathbf{r}}}{2m} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2}\sum_{I\neq J}\frac{Z_I Z_J e^2}{|\mathbf{R}_I - \mathbf{R}_J|} - \sum_{I,i}\frac{Z_I e^2}{|\mathbf{r}_i - \mathbf{R}_I|}, \tag{4}$$

In which capital letter, I and J are atomic ions, and i and j are electron indices.

2.2.1 Born-Oppenheimer Approximation

We are doing the following ansatz for the wavefunction:

$$\Psi(\{r\},\{R\}) = \Phi(\{r\},\{R\})\chi(\{R\}), \tag{5}$$

In which, we move all the *strong depend* of $\{R\}$ into $\chi(\{R\})$ and vice versa. So, the ion kinetic operator acts on the total wave function:

$$T_{\text{nuclei}}\Psi = -\frac{\hbar^2}{2m_I}\sum_I \nabla^2_{R_I}(\Phi\chi) = -\frac{\hbar^2}{2m}[\nabla^2_{R_I}(\Phi)\chi + 2(\nabla_{R_I}\Phi)(\nabla_{R_I}\chi) + \nabla^2_{R_I}(\chi)\Phi] \tag{6}$$

Since the derivation of the "strong function" $\chi(\{R\})$ will dominate the rest of this term, we can simply ignore it. Another result of this ansatz is:

$$\int |\chi(\{\mathbf{r}\},\{\mathbf{R}\})\Phi(\{\mathbf{R}\})|^2 d\{\mathbf{r}\}d\{\mathbf{R}\} = \int |\chi|^2 d\mathbf{r} = 1\,\forall R \tag{7}$$

Including these discussions in the Schrödinger equation, multiply on the left side by the Φ and take the integral over the $\mathbf{r}$:

$$H = T_N + H_e + H_{Ie} + H_{\text{Ion-Ion}}$$

$$\int \bar{\Phi} H \Phi \chi d\mathbf{r} = (T_{\textbf{nuclei}} + H_{\text{Ion-Ion}} + U(\{\mathbf{R}\}))\chi = E\chi, \tag{8}$$

$$U(\{\mathbf{R}\}) = \int \Phi(\{r\},\{R\})\Big(H_e + H_{Ie}\Big) \tag{9}$$

The (8) is called the Schrödinger equation for ions, and when we define the (9), we also define the Schrödinger equation for the electron:

$$\Bigg(-\frac{\hbar^2}{2m}\sum_i \nabla_i^2 + \sum_{i\neq j}\frac{1}{2}\frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i,I}\frac{e^2 Z_I}{\mathbf{r}_i - \mathbf{R}_j}\Bigg)\Phi(\{\mathbf{r}\},\{\mathbf{R}\}) = E\Phi(\{\mathbf{r}\},\{\mathbf{R}\}) \tag{10}$$

Keep in mind that the $\Phi(\{\mathbf{r}\},\{\mathbf{R}\})$ weakly depend on $\{\mathbf{R}\}$, therefore we can safely omit this notation, resulting in a familiar form:

$$\Bigg(-\frac{\hbar^2}{2m}\sum_i \nabla_i^2 + \sum_{i\neq j}\frac{1}{2}\frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i,I}\frac{e^2 Z_I}{\mathbf{r}_i - \mathbf{R}_j}\Bigg)\Phi(\{\mathbf{r}\}) = E\Phi(\{\mathbf{r}\}) \tag{11}$$

2.2.2 Ab-initio Approach

The ab-initio calculation aiming to solve the Schrödinger equation $H\Psi = E\Psi$ for a many-body solid state system consisting of electrons and nuclei is showing to be impossible. To diagonalise this kind of system, many approximations and numerical techniques have been developed. One of these methods is called Density Functional Theory (DFT), which has been widely used for its effective computational cost and precision of describing a wide number of problems in material science, in particular and condensed matter physics in general. In this section, we will shortly describe the basics of DFT.

Assuming that the system has electronic density $n(\mathbf{r})$, then the total energy can be described as a functional of this density:

$$E[n] = T[n] + E_{ei}[n] + E_H[n] + E_{xc}[n] + E_{ii}[n]. \tag{12}$$

Where each term in the equation above indicates kinetic, electron-ion interaction, Hartree energies, exchange energy and ion-ion interaction, respectively. As we discussed in the previous section, the Born-Oppenheimer approximation allows us to decouple the Ion and electron equations into two "separate" problems, which means that the nuclei' energy $E_{ii}[n]$ will be treated as a constant, and can be neglected by shifting the ground energy.

Hohenberg-Kohn theorem (1964)

If we assuming that the electron system with external potential $V_{ext}(\mathbf{r})$ have then density $n(\mathbf{r})$ as the ground state, then can there exist another $V_{ext\,2}(\mathbf{r})$ that also have $n(\mathbf{r})$ as the ground state?
The Hohenberg-Kohn theorem says:

"No, you can't!"

Hohenberg-Kohn theorem "not" said that, at least this short
:v.

Thus, the external potential is uniquely functional of the density corresponding to its true ground state. Then, we can minimise the functional to find the true ground state density[2].

DFT utilises the single particle wave function of an electron instead of the many-body wavefunction to determine the energy of the ground state. By knowing the form of $E[n]$, we can minimise it to find the ground state, but usually this true form density is not explicitly known.
Kohn and Sham showed that the density of interacting electrons in an external potential is equivalent to that of non-interacting electrons in an effective potential. This effective potential include the external potential V_{ext}, Hatree energy $V_H = \int d^3\mathbf{r}' \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}$ and exchange correlation $V_{xc} = \frac{\delta E_{ex}[n(\mathbf{r})]}{\delta n(\mathbf{r})}$.

This so-called exchange correlation captures the difference in energy between the interacting and non-interacting electron system. This function maintains the exchange that comes from the Pauli exclusion principle and dynamic correlation. It incorporates the corrections that come from the kinetic energy of the interacting system, and the self-error correlation comes from the Hartree potential. In short, it contains all the correlations for all the approximations that we will apply to other components. Therefore, it is very complicated to find the correct function for the real system, leading to many approximations, which will be discussed in the next section.
With these, the true density can be calculated as the solution of a set of single-particle Schrödinger-like equations called Kohn-Sham (KS) equations:

$$(T + V_{ext} + V_H + V_{xc})\varphi_i(\mathbf{r}) = \varepsilon_i \varphi_i(\mathbf{r}), \tag{13}$$

in which the eigen-energy ε_i corresponding to the singe-particle eigen-function φ_i. The single-particle density will be the sum over the occupied states:

$$\rho(\mathbf{r}) = \sum_{\text{occ.}} |\varphi_i(\mathbf{r})|^2. \tag{14}$$

Since the Hamiltonian operator in (13) also depends on the density $n(\mathbf{r})$, the iteration solving process is required for this eigenvalue problem. Last but not least, it is worth keeping in mind that the eigen-energy ε_i does not alone give the energy of the system, but its eigenvector and itself contribute to the total energy:

$$E[n] = E_{ii}[n] + \sum_{\text{occ.}} \varepsilon_i + E_{xc}[n] - \int d\mathbf{r}\, n(\mathbf{r})(V_{xc} + \frac{1}{2} V_H(\mathbf{r})). \tag{15}$$

2.2.3 Approximating The Exchange Functional

In principle, DFT give exact values, but the unknown exchange functional $E_{xc}[n]$ is calculated with various approaches using corresponding approximations. The simplest approximation is the Local Density Approximation (LDA), in which the exchange functional comes from a homogeneous gas. Since $V_{xc} = \frac{\delta E_{xc}[n]}{\delta n}$ and $n(\mathbf{r}) = \text{const}$ in the homogeneous electron gas, $V_{xc}(\mathbf{r}) = V_{xc}[\rho(\mathbf{r})]$[2]. But for the case of a non-homogeneous electron gas, the further derivative orders will be included in the total functional:

$$V_{xc}[n](\mathbf{r}) = V_{xc}[n(\mathbf{r}), \nabla n(\mathbf{r}), \nabla^2 n(\mathbf{r}), ...] \tag{16}$$

The next approximation includes the first derivative of density, therefore being called "Generalised Gradient Approximation (GGA)"[3–5]. This is useful for systems with the slow variation of densities.

2.2.4 Basis sets and plane waves

The Kohn-Sham orbital function can be expanded via a set of basis functions of plane waves:

$$\varphi_i(\mathbf{r}) = \sum_{\alpha} c_{i\alpha}\phi_\alpha(\mathbf{r}), \tag{17}$$

where $c_{i\alpha}$ is the coefficient and ϕ_α is the basis function.

As discussed above, the wave function of a particle inside the periodic potential has the plane-wave term, showing it is a good choice to be a basic set. Using this basis set will require a sum over an infinite number of values of $\mathbf{G}$ for each point in the k-space. But, Bloch functions have low energy have more contribute than the high value of $\mathbf{k}+\mathbf{G}$, therefore it is sufficient to introduce the energy cutoff E_{cut} to only include the energy of plane wave less than $E_{cut} \sim \mathbf{G}^2_{cut}$. Therefore, we have the combination of the KS function:

$$\varphi_{\mathbf{k}}(\mathbf{r}) = \sum_{|\mathbf{G}+\mathbf{K}|<\mathbf{G}_{cut}} c_{\mathbf{k}+\mathbf{G}} e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}. \tag{18}$$

The plane-wave basis set is utilised in an integrated suite of computer codes for electronic structure calculations and materials modelling, called Quantum ESPRESSO[6–8], which we use in this thesis.

2.2.5 Pseudopotentials

The electrons bound to an atom will have a high oscillating wave function near the core. If we include these oscillations in our calculation, the basic set has to be large enough to capture these oscillations, but it is not very important in the range of inter-atom distances. To minimise the computational cost for this process, the Pseudopotential uses the frozen core approximation, which replaces the density of core electrons with a smoothed density that matches the properties of true ion cores. This argument is supported by the fact that in the material required, the bonding between atoms, these bonds are mostly constructed by the valence electron, leaving the core electron wave function unchanged.

The relativistic effect, both scalar and fully one, will also contribute to these pseudopotentials. In scalar relativistic approximations, relativistic effects such as mass-velocity, Darwin, and other higher-order corrections except spin-orbit coupling (SOC) are taken into account, whereas fully relativistic treatments also include SOC. With the software we are using, Quantum Espresso supports projector-augmented wave (PAW) methods, ultrasoft pseudopotentials, and norm-conserving pseudopotentials.

2.3 Wannier Function and Wannierization Process

2.3.1 One-band Wannier functions

There are many ways to choose the orthornormal basis, one of them is the Wannier function. From the wavefunction according to the Bloch theorem:

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}) \tag{19}$$

We can define the Wannier function by taking the inverse Fourier transform:

$$\Phi_{\mathbf{R}}(\mathbf{r}) = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} \psi_{\mathbf{k}}(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{R}} = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} u_{\mathbf{k}}(\mathbf{r}) e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R})} = \Phi(\mathbf{r}-\mathbf{R}) \tag{20}$$

From that definition, the Wannier function has some important properties:

- For any lattice vetor $\mathbf{R}'$:

$$\Phi_{\mathbf{R}+\mathbf{R}'}(\mathbf{r}+\mathbf{R}') \propto u_{\mathbf{k}}(\mathbf{r}+\mathbf{R}') e^{i\mathbf{k}\cdot[(\mathbf{r}+\mathbf{R}')-(\mathbf{R}+\mathbf{R}')]} = u_{\mathbf{k}}(\mathbf{r}) e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R})} = \Phi_{\mathbf{R}}(\mathbf{r}) \tag{21}$$

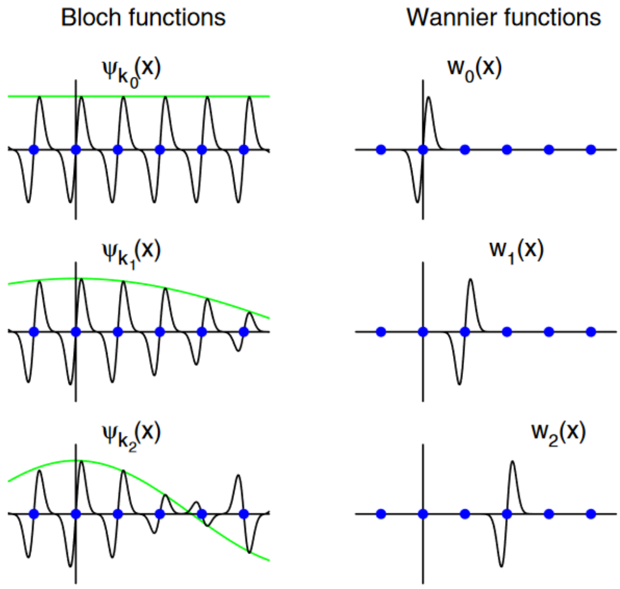


Figure 3: Comparison between Bloch and Wannier function. The Bloch function oscillates all over the plane while the Wannier function is localised near the lattice point

- Orthonormal (also one of the reasons why we choose it):

$$\begin{aligned}\int_V \Phi^*_{\mathbf{R}}(\mathbf{r})\Phi_{\mathbf{R'}}(\mathbf{r})d\mathbf{r} =& \frac{1}{N}\sum_{\mathbf{k},\mathbf{k'}}\int_V u^*_{\mathbf{k}}(\mathbf{r})u_{\mathbf{k'}}(\mathbf{r})e^{-i\mathbf{kr}}e^{i\mathbf{k'r}}d\mathbf{r}e^{i\mathbf{kR}}e^{-i\mathbf{k'R'}} \\ =& \frac{1}{N}\sum_{\mathbf{k},\mathbf{k'}}\delta_{\mathbf{kk'}}e^{i\mathbf{k}\cdot\mathbf{R}-\mathbf{k'}\cdot\mathbf{R'}} = \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{R}-\mathbf{R'})} = \delta_{\mathbf{RR'}}\end{aligned} \tag{22}$$

- Localise around the atom:

$$\int_V \Phi^*_{\mathbf{R}}(\mathbf{r})\Phi_{\mathbf{R}}(\mathbf{r}+\mathbf{R'})d\mathbf{r} = \int_V \Phi^*_{\mathbf{R}}(\mathbf{r})\Phi_{\mathbf{R}-\mathbf{R'}}(\mathbf{r})d\mathbf{r} = \sum_k e^{i\mathbf{k}\cdot\mathbf{R'}} = 0 \quad (\mathbf{R'} \neq 0, \mathbf{k} \neq 0) \tag{23}$$

2.3.2 Multi-band

Gauge freedom:

If we include the phase into the wavefunction:

$$\left|\tilde{\psi}_n(\mathbf{r})\right\rangle = e^{i\varphi_{n\mathbf{k}}}\left|\psi_n(\mathbf{k})\right\rangle, \tag{24}$$

without changing the physical description of the system[9], in which $\varphi(n, \mathbf{k})$ periodic in k-space. The choice of $\nabla u(\mathbf{n,k})$ for it to be well defined on all points is called "smooth gauge". There are many ways to choose the "smooth gauge" for each Bloch periodic function, so that the Wannier function will be well localised, resulting in the "non-unique" nature of it. Those results are piping to the "even more non-unique" properties of Wannier functions since each set of phases can result in different shapes and spreads of Wannier functions and vice versa. The Schrödinger isn't in favour of any of the gauge choices, therefore, the non-uniqueness is unavoidable.

Multiband case:

In the solid-state, there are many cases where not only one band but several will degenerate and cross among themselves. In case the J bands are isolated from the other bands by a gap of energy above and below, we can define a unitary transformation for this manifold:

$$\left|\tilde{\psi}_n(\mathbf{k})\right\rangle = \sum_m U_{mn}(\mathbf{k}) \left|\psi_m(\mathbf{k})\right\rangle, \tag{25}$$

In which the $U_{mn}(\mathbf{k})$ periodic in k-space. This transformation proves to be especially useful when describing the degenerate case (for example, the 3p-degeneracy at Gamma of Si bandstructure). In short, this rotation is chosen to "cancel out" the discontinuities of $\psi_n(\mathbf{k})$ that are *per se* [2] not smooth, in such a way that smoothness is restored, i.e., that the $\nabla\tilde{\psi}_n(\mathbf{k})$ is regular at all k point. These arguments allow us to introduce the definition of the Wannier function from the Bloch function:

$$|\mathbf{R}, n\rangle = \frac{(2\pi)^3}{V} \int d\mathbf{k} \sum_m^J U_{mn}(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{R}} \left|\psi_{m\mathbf{k}}\right\rangle. \tag{26}$$

By adopting the convention:

$$\left\langle n, \mathbf{k}\middle|m, \mathbf{k}'\right\rangle = \frac{(2\pi)^3}{V}\delta_{nm}\delta(\mathbf{k} - k'),$$

The Wannier function in (26) will be normalized[3]:

$$\langle \mathbf{R}n|\mathbf{R}'m\rangle = \delta_{nm}\delta_{\mathbf{R}\mathbf{R}'}.$$

For the convenience of numerical solving later on, we can pass it into the discrete k-space through the convention:

$$\left|\psi_n(\mathbf{k})\right\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\cdot\mathbf{R}} \left|n\, \mathbf{R}\right\rangle, \tag{27}$$

$$\left|n\, \mathbf{R}\right\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}} \left|\psi_n(\mathbf{k})\right\rangle, \tag{28}$$

In which $\left\langle n, \mathbf{k}\middle|m, \mathbf{k}'\right\rangle = \delta_{nm}\delta_{\mathbf{k}\mathbf{k}'}$, give us the projection operator that operates on two bases:

$$P = \sum_{n\mathbf{k}} |n\mathbf{k}\rangle\langle n\mathbf{k}| = \sum_{n\mathbf{R}} |\mathbf{R}n\rangle\langle\mathbf{R}n|. \tag{29}$$

2.3.3 Maximally localised Wannier functions (MLWF)

So, up to now, we know that the gauge choice can affect the centre and the spread of the Wannier function. Our task from now on is to find a correct mixing matrix $U_{mn}^{(\mathbf{k})}$ to satisfy some criterion. In here, we will use the criterion developed by Marzari and Vanderbilt [10], considering the spread of the Wannier function in their home cell:

$$\Omega = \sum_n \langle \mathbf{0}\, n| (r - \bar{r})^2 |\mathbf{0}\, n\rangle = \sum_n (\langle \mathbf{0}\, n| r^2 |\mathbf{0}\, n\rangle - \langle \mathbf{0}\, n| r |\mathbf{0}\, n\rangle^2), \tag{30}$$

or,

$$\Omega = \sum_n (\langle \mathbf{0}\, n| r^2 |\mathbf{0}\, n\rangle - \sum_{m\mathbf{R}} \langle \mathbf{R}\, m| r |\mathbf{0}\, n\rangle^2 + \sum_{n\mathbf{R}\neq m\mathbf{0}} \langle \mathbf{R}\, m| r |\mathbf{0}\, n\rangle^2) = \Omega_I + \tilde{\Omega}, \tag{31}$$

[2] itself

[3] Actually, the Wannier function is not entirely localised in one cell but periodic at the supercell scale, the process later on is finding the maximally localised state in one cell.

in which,

$$\Omega_I = \sum_n (\langle \mathbf{0}\, n | \, r^2 \, | \mathbf{0}\, n \rangle - \sum_{m\mathbf{R}} \langle \mathbf{R}\, m | \, r \, | \mathbf{0}\, n \rangle^2), \tag{32}$$

$$\tilde{\Omega} = \sum_{n\mathbf{R} \neq m\mathbf{0}} \langle \mathbf{R}\, m | \, r \, | \mathbf{0}\, n \rangle^2 . \tag{33}$$

Eq. (32) is *gauge invariant* under the transformation in eq. (25)[9;11], while eq. (33) is not. So, the job is to minimise the last term by choosing the correct $U_{mn}^{(\mathbf{k})}$, therefore minimise the functional Ω. Therefore, by doing the Wannierization step, we also project the Hamiltonian on a Wannier basis composed of the most contributing orbital of the bands we are considering, which allows us to further calculate the other properties of the material.

2.4 Berry Phase

Imagine we have a system that has a parameter that can be tuned with time, for example, the electric field $\mathbf{E}(t)$ varies in time. If the system from the beginning stays in the ground state and the variation of the parameter is slow enough not to excite the system to another state or experience the degeneracy of the band structure anywhere, it will stay in the corresponding eigenstate at the end of the time evolution. This is called the "Adiabatic Evolution", or in the original context of Born and Fock:

> "A physical system remains in its instantaneous eigenstate if a given perturbation is acting on it slowly enough and if there is a gap between the eigenvalue and the rest of the Hamiltonian's spectrum."[12]

Keep this idea in mind, we can express the wave function in the form[13]:

$$|\Psi_m(t)\rangle = e^{-\frac{i}{\hbar}\int_0^t E(\mathbf{R}(t'))dt'} e^{i\gamma_m} |m(\mathbf{R}(t))\rangle , \tag{34}$$

In which the vector $\mathbf{R}(t)$ showing there value of parameter and it evolution in the parameter space, the ket $|m(\mathbf{R}(t))\rangle$, showing the system staying in the same eigenstate in respect to the parameter vector $\mathbf{R}(t)$. The geometry phase γ_m is purely imaginary, therefore the $i\gamma_m$ have the real results, its value can be calculated through:

$$\gamma_m = i\int_0^t \langle m(\mathbf{R}(t)) | \dot{m}(\mathbf{R}(t))\rangle \, dt' = i\int \langle m(\mathbf{R}(t))| \, \nabla_{\mathbf{R}} \, |m(\mathbf{R}(t))\rangle \, d\mathbf{R} = i\int \mathcal{A}_{\mathbf{R}} d\mathbf{R}, \tag{35}$$

In which the geometry phase clearly does not depend on the time we vary the parameter, but on the path of the variable in its own space. If at the end of variation, the system goes back to its initial condition, then according to Stock's theorem:

$$\gamma_m = i\oint \langle m(\mathbf{R}(t))| \, \nabla_{\mathbf{R}} \, |m(\mathbf{R}(t))\rangle \, d\mathbf{R} = i\iint_S F_{\mu\nu} dS^{\mu\nu}, \tag{36}$$

In which:

$$F_{\mu\nu} = \frac{\partial \mathcal{A}_i}{\partial R_j} - \frac{\partial \mathcal{A}_j}{\partial R_i}. \tag{37}$$

The $F_{\mu\nu}$ is called *Berry curvature* and $S^{\mu\nu}$ is the physical meaning of the area corresponding to the change of parameters in it-space.

3 Methodology

The structure, electronic band structure properties of RuO_2 will be shown, and are calculated by DFT as implemented in Quantum Espresso 7.2, projection, wannierization and other properties were calculated on Wannier90 and its utility Postwannier. These programs are installed on the hardware of Leonardo's computing node from the Cineca cluster. All the plotting will be done on Gnuplot or Python libraries.

3.1 DFT calculations

Due to the strong correlation of this system, generalised gradient approximation (GGA) of the exchange-correlation functional was used with the Perdew-Burke-Ernzerhof (PBE) functionals and the scalar relativistic projector-augmented wave (PAW) method for the pseudopotentials from the PPs library of QE website for both Ru and O atoms.
In this work, we only work with the scalar case with the use of plane-wave basis with an energy cutoff of 90 Ry for the wave function and 700 Ry for the density. The Brillouin zone was sampled using a Γ-center Monkhorst-Pack $10 \times 10 \times 16$. The convergence threshold was set to 10^{-10} Ry. As mentioned about the properties of antiferromagnetism, we need the structure of the antiferromagnetic material for the calculation. Therefore, the unit cell we define will come with 2 Ru atoms and 4 O atoms (as shown in 2), each Ru atom will be assumed to start with the magnetic moment of 0.8 μ_B with opposite direction, create the Néel direction along the $[001]$[14].

The equilibrium structure is obtained by relax the atomic coordinates and lattice structure parameter with a force threshold of 10^{-3} Ry/Bohr.

Due to the strong correlation of the d-orbital electrons of Ru atom, we need to using the GGA + U approach[15] with the Hubbard parameter U to be 1.5 eV[14].

3.2 Projection and Wannierization

After the DFT calculation, we project the band structure onto the band to confirm the number of Wannier functions that have to be used in order to sufficiently describe the system. We also have to narrow down the window of the projection according to the contribution of the atomic orbital onto the band structure, rather than working over the entire band spectrum.

3.3 Berry Curvature

The Berry curvature $\Omega_k = \varepsilon^{ijk}\Omega_{ij}$, in which ε^{ijk} is the Levi-Civita notation, will be calculated through Postwannier by the equation:

$$\begin{aligned}\Omega_{\alpha\beta}(\mathbf{k}) &= -2\sum_n \mathrm{Im}\left\langle \nabla_{k_\alpha} u_{n\mathbf{k}} \middle| \nabla_{k_\beta} u_{n\mathbf{k}} \right\rangle \\ &= i \sum_{n,n'\neq n} \frac{1}{(\varepsilon_n(\mathbf{k}) - \varepsilon_{n'}(\mathbf{k}))^2} \left(\langle n| \frac{\partial H}{\partial k_\alpha} |n'\rangle \langle n'| \frac{\partial H}{\partial k_\beta} |n\rangle - \langle n| \frac{\partial H}{\partial k_\beta} |n'\rangle \langle n'| \frac{\partial H}{\partial k_\alpha} |n\rangle \right) \end{aligned} \tag{38}$$

4 Results and Discussion

4.1 Lattice's Structure

Starting from the tetragonal lattice configuration, we allow the lattice to relax without changing the symmetry of the lattice. Using the larger value of wavefunction-cutoff 90 Ry and the scf convergence threshold 5.0E−12, we achieve $a = 4.5405$ Å and $c = 3.1323$ as the lattice parameter. Our results show good agreement with other theory calculations and experimental data[16–20].

	a	c	Max error (in percent)
Experiment	4.4994[16]	3.1071[16]	0.91
	4.491[17]	3.106[17]	1.09
	4.49[18]	3.104[18]	1.11
Theory	4.561[19]	3.102[19]	0.97
	4.5[20]	3.12[20]	0.89

4.2 Spin-splitting Electronic structure

In this part, we will show that by choosing a difference k-path than the original ansatz[21], the properties of the RuO_2 in splitting the bandstructure will be shown much more clearly.

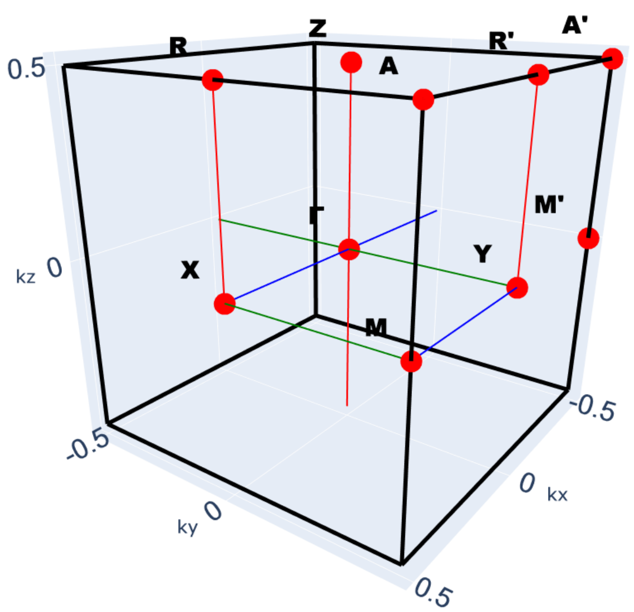


Figure 4: Brillouin Zone of RuO_2

The 3D Brillouin zone with the high symmetry points are being shown in 4, in which the original path contains: $\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma \rightarrow Z \rightarrow R \rightarrow A \rightarrow Z|X \rightarrow R|M \rightarrow A$ have the discontinuing on the k-path and not showing the rotational symmetry of the alternative spin-spillting bandstructure on the diagonal of the plane perpendicular to $[001]$. Therefore, we suggest another path that contains it:

$$R \rightarrow Y \rightarrow \Gamma \rightarrow X \rightarrow M \rightarrow \Gamma \rightarrow M' \rightarrow A' \rightarrow Z \rightarrow R' \rightarrow A$$

Pristine	Ru1	Ru2	absolute	Fermi
	0.7149	-0.7161	2.21	12.2534

Table 1: The magnetisation moment of 2 Ru atoms, absolute magnetisation per unit cell and Fermi level of RuO_2. Ru1 and Ru2 indicate the magnetisation moment of major and minor spins in μ_B per unit cell, respectively. The absolute value showing the absolute magnetisation in the same unit, the Fermi level is in eV.

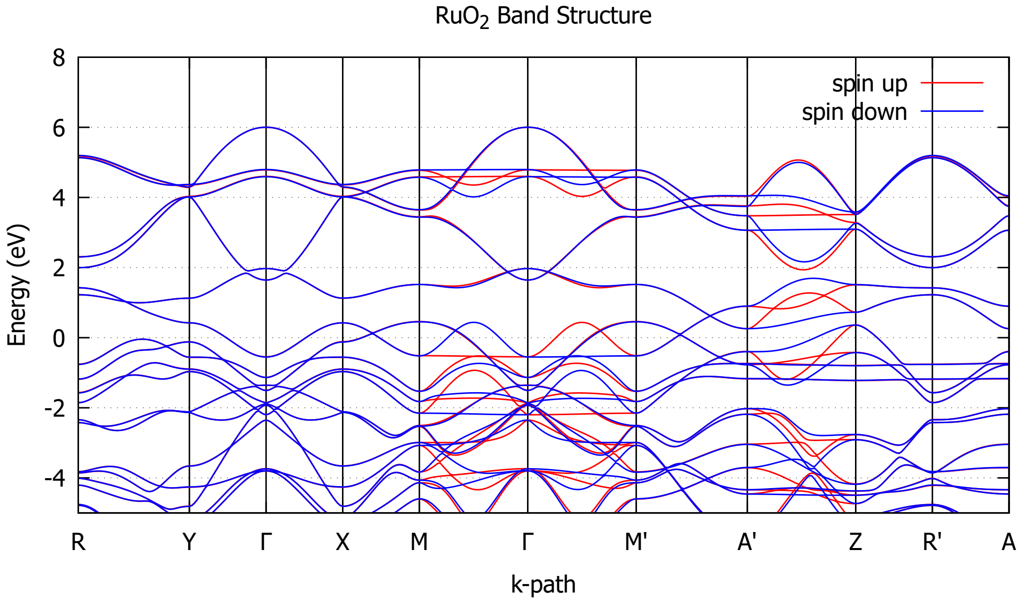


Figure 5: Calculated spin-polarized band structure from DFT+U calculations with U = 1.5 eV

As shown in Fig. 5, the diagonal path contains the spin-splitting band structure to be around 100s of meV near the Fermi level, confirming the existence of spin-splitting without the effect of SOC. The changing of spin up to be down, and vice versa, connect through the rotational symmetry C_{4z} of the band structure, corresponding to the idea of the symmetry of the d-orbital wavefunction-like of the spin wave in the k space.

Also from the band structure, we observe the rotational symmetry of the system, corresponding to the same in real lattice, when we have to take 90 90-degree rotation of the system after the translation between two Ru atoms in the unit cell, followed by the spin flip to recover the same system. The spin up will change their position with spin down under the 90 degree rotation (ΓM to $\Gamma M'$ and vice versa. If we take the second rotation and do another spin flip, we will have the same band structure under 180-degree rotation, or $\varepsilon_{\uparrow}(\mathbf{k}) = \varepsilon_{\uparrow}(-\mathbf{k}) \neq \varepsilon_{\downarrow}(-\mathbf{k})$, indicating the breaking of time-reversal symmetry properties of this system. Another observation on the same path is the flat-band, indicating the very well localised state and or particular, d-orbital state of the Ru atom. This argument can be supported by the projection of each d-type orbital on the band structure in the appendix.

4.3 Effect of Doping The System

To see the effect of doping the system, we do the DFT calculation under the same conditions of lattice structure and other parameters, the results are shown in the table 2. The electron doping causes a significant decrease in the on-site magnetisation of each Ru atom and therefore the absolute magnetisation (reduced at most $19\,\%$ at the doping of 0.5 electron per unit cell). We also observe a greatly increased Fermi level (300 meV) for the case, respectively. While, the doping of hole into the system on the other hand, show the stable of the magnetization of the magnetization moment but a small reduce in the Fermi level.

My explanation for these difference in changing of the magnetization can come from the ratio and the spin-spillting, particular on the path ΓM or $\Gamma M'$ in the band structure as shown in Fig. 5. Let us make everything easier to see by only considering the absolute magnetisation of the unit cell or in the entire Brillouin zone. The initial ground state contain one spin channel (the line almost straight in the ΓM path, therefore, any increasing of the Fermi level will "weaken" this spin polarized at this specific k-point (if two spin channels of a band at a k-point are both under the Fermi level, it will be 0 for it contribution into the absolute magnetization of the unit cell). The $A'Z$ path will not be taken into account much in this effect due to the equivalence between the two spin channels at the Fermi level. On the other hand, the non-reduction of the magnetization moment of Ru and absolute and the low decrease of the Fermi level come from the high density of states on the $RY\Gamma$ path, which only contribute their density of states when the Fermi level decrease but not vice versa, therefore further increase the ratio between the spin-splitting state and the spin-degeneracy state.

Following the same idea, I also have two predictions for this system. If the doping of the system does not change much the band structure and we continue doping electrons to the system, it will eventually fill all the spin channels on the ΓM path and become neutral in magnetisation. On the other hand, the opposite case will be harder to achieve when we try to reduce the Fermi level, the contribution of the $RY\Gamma$ bands, as mentioned above. But if we can reduce the Fermi level down by 0.5 eV, we will observe a sudden collapse of the magnetisation when it passes the one-spin channel flat bands. On the other hand, there is also argue about the exist of altermagnetic state and on-site magnetization at low U (smaller than 1 eV[14], in particular), which I suggest come by the reduction of the γM spin-hill below the Fermi level, it make sense because the Hubbard parameter indicate the repulsive of the spin channel, the weaker it is, the smaller the gap, leave the Fermi level do the rest. My argument for the band structure on the ΓM under the variation of the U parameter will be that the hill will decrease in height and the ground will become curved due to the high density of states of the ground.

Dope per unit cell	Hole				electron			
	Ru1	Ru2	absolute	Fermi level	Ru1	Ru2	absolute	Fermi level
0.1	0.7165	-0.7185	2.2	12.1695	0.7143	-0.7153	2.21	12.3713
0.2	0.719	-0.7222	2.2	12.1078	0.6807	-0.6815	2.11	12.4585
0.3	0.7182	-0.7233	2.19	12.0532	0.669	-0.6706	2.08	12.564
0.4	0.7134	-0.7204	2.17	12.0008	0.6256	-0.6268	1.94	12.6131
0.5	0.713	-0.7213	2.17	11.9491	0.579	-0.5809	1.8	12.657

Table 2: Comparison of doping the system, the format is the same as Fig. 1

4.4 Orbitally-projected Bandstructure

After confirming the spin-splitting band structure, we project it onto the orbital of each atom (d- and p-orbital for Ru and O, respectively) through the utility projwfc.x of Quantum Espresso. We focus the projection on the bands near the Fermi level, which play the key role for metal materials. The Fig. 6 above comes from the relative density of states at each point and spin, and then being summed over. A more particular showing of projection for each band will be given in the appendix with a further discussion of the symmetry corresponding between the system and orbitals. Other research focuses in a particular set of Ru d-orbital call $t_{2g} = \{d_{xy}, d_{yz}, d_{zx}\}$[14], but as we discussed in the theory of Wannierization process, the isolated bands have to be projected on all the contributor (the contribution of 2 other d-orbital can be seen in the appendix also) for the justify reason. (The choice of t_{2g} actually also can be done by narrowing down the froze window (not the forcus window) of the Wannier90 code to be around the Fermi level, which will freeze other band and only forcus on the band structure near the Fermi surface and high symmetry point, this can be confirm from our calculation of the total projection of Bandstructure onto this group nearly match the number of orbital (which is 3). Even this choice look like it protect the symmetry of D_{4h} symmetry of RuO_2, it can lead to the error when expand through out the Brillouin zone due to the contribution of E_{2g} (contains two other d-orbital) group being neglected.)

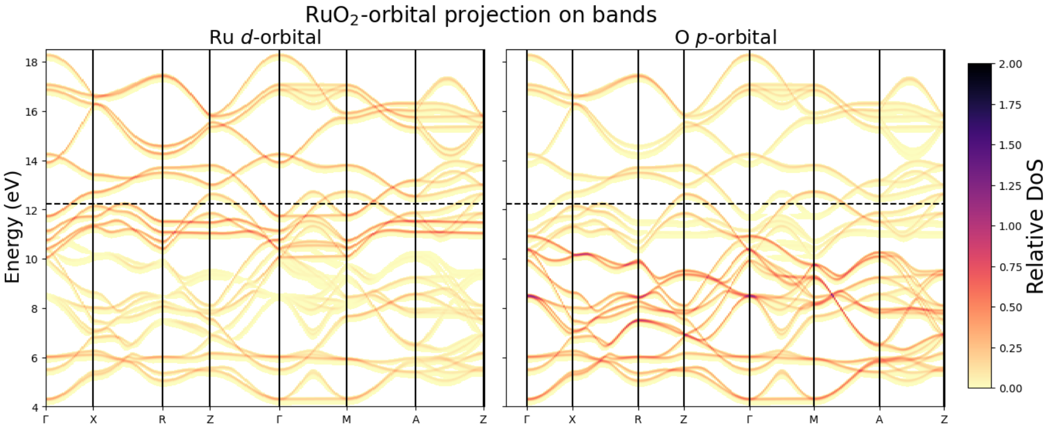


Figure 6: The contribution of Ru-d (left) and O-p (right) orbitals into bandstructure (both spin)

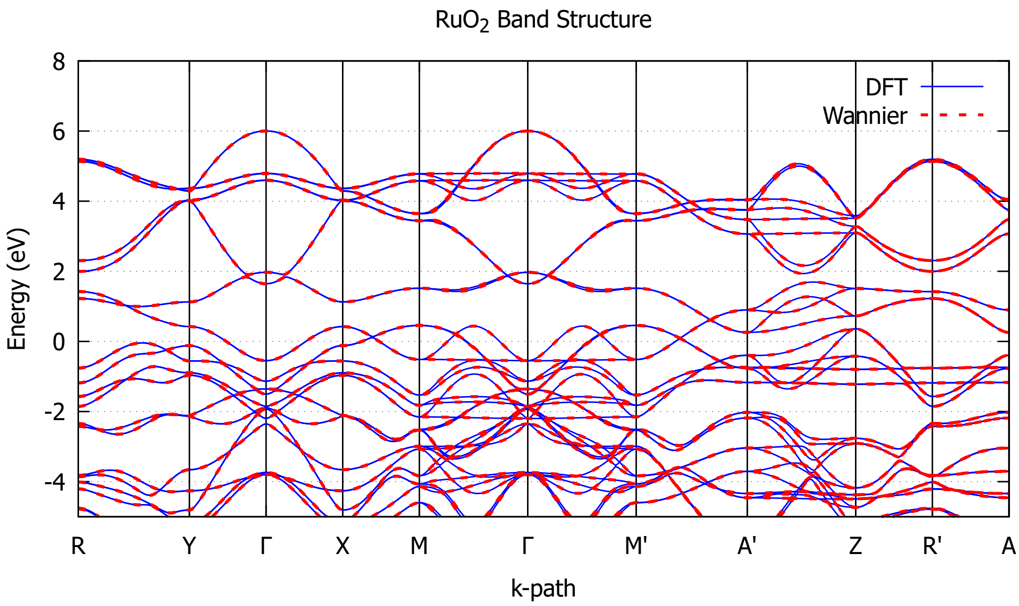


Figure 7: Comparison band structure between DFT and Wannierization

4.5 Wannierization

Therefore, we will need all Ru-d and O-p orbitals in the projection process. With a total of 22 orbitals, we can do the projection onto the Wannier basis. By setting the window of projection to be between $(3, 20)$ eV, this metallic system has the contribution from each band spread throughout the system rather than focusing in some zone; therefore, we are not using any of the freeze window in this calculation. The number of steps for the wannierization process will be chosen to be 200, after confirming that this number is sufficient to converge the total spread of the wannier function.

After the wannierization, its band structure has been confirmed to match the Quantum Espresso results, as shown in Fig. 7. The Fermi surface of RuO_2 has also been plotted in , showing the non-degenerate band structure. The spin polarisation connected through rotation and spin flip has also been confirmed again in this figure. Also need to notice the candy shape in the middle of the Brillouin zone, which is showing the closed Fermi surface for this band and its anisotropic conductivity of each spin channel. The other band also closed, not in the first Brillouin zone, but on the overall k-plane instead. Closed band, showing that these electrons are well localised in the k-space, allowing them to contribute a finite density of carriers available for conduction. All those pockets show this is a good metal since it contains a high density of conduction electrons.

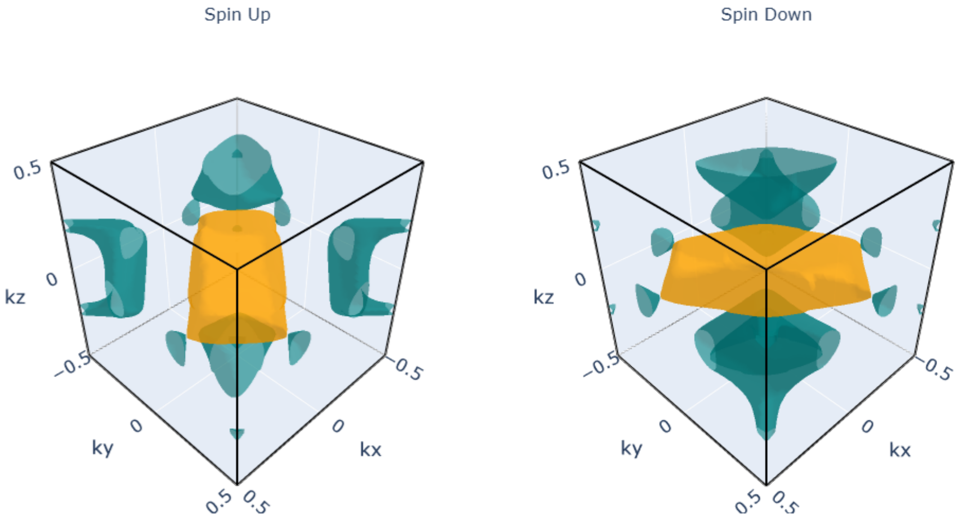


Figure 8: Fermi surface for both spins. Two different colours show different bands crossing the Fermi level

4.6 Choosing k-slice

So, we mentioned the time reversal symmetry breaking of this system, we will now ask the question about its conductivity properties. But we keep in mind the idea that this is a 3-D Brillouin zone, and therefore, we need to cut through the zone in order to examine its properties. According to the band structure in Fig. 5, we see that the ΓAM should be a good plane to be cut since it contains the Dirac point near the Fermi level (about 100-200 meV), which can be discovered by experiments. Combined with the symmetry properties of this system, we should cut through the entire Brillouin zone rather than just a quarter of it, as shown in Fig. 9.

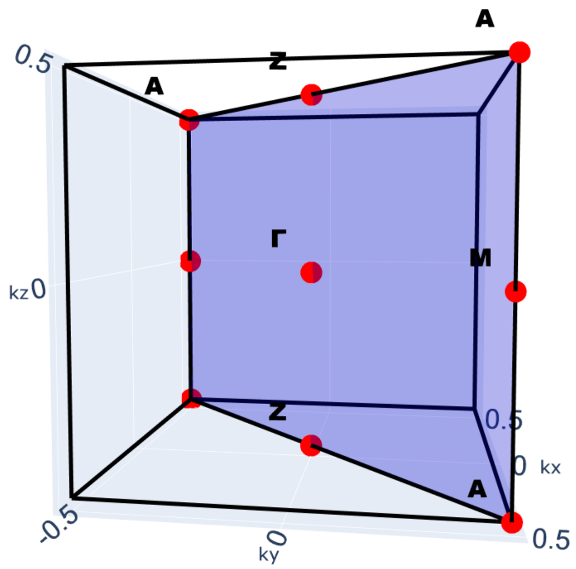
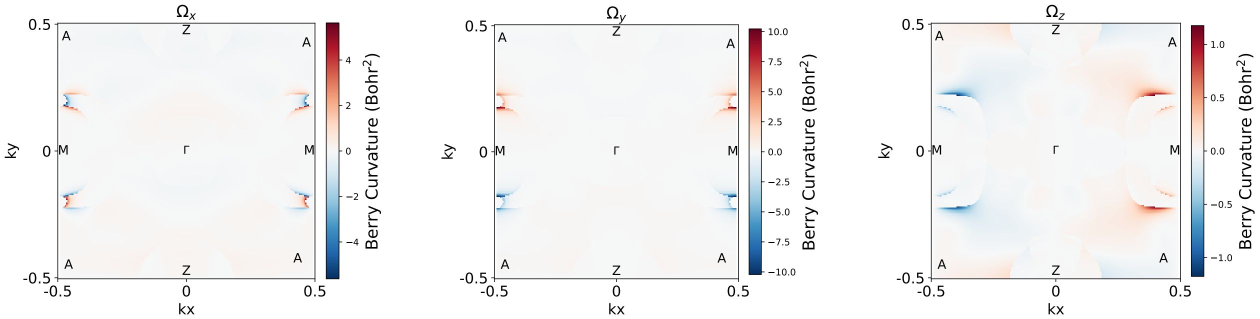


Figure 9: The k-slice going through ΓAM points



(a) x component (b) y component (c) z component

Figure 10: Berry curvature

4.7 Berry Curvature

Using this plane, we calculated and plotted the Berry curvature with k-mesh $50 \times 50 \times 50$. As shown in Fig. 10, we see the trace of the Dirac point on the AM path, which has a semi-circle shape, indicating the closing of the Dirac point. An interesting aspect of these figures comes from the shape of the Fermi surface in Fig. 4.5, in the x-component (Fig. 10a) and y-component (Fig. 10b), we clearly see two different shapes, the semi-circle mentioned above and two more lines showing the trace of the hand-holding, which indicate two separate spin-polarisation Fermi surfaces. These two shapes meet at AM direction, showing the degeneration of two spins on the ΓZ direction. Those not very clear on the other two components, the z component has two arcs at the Z point, showing two spin Fermi surfaces.

4.8 Next steps

From these calculations, we suggest the next step will be to evaluate the conductivity of the system. The Berry curvature of the pristine case is still low and gives the cancellation when taking the summation over the Brillouin zone. We can break the inversion symmetry by placing the system under the effect of strain, putting on the substrate, replacing some atom by the same kind of element (for example, Ru replaced with Cr for some ratio [14]), introducing imputrity in the superlattice or breaking the time-reversal symmetry by including the spin-orbit coupling, applying the external field. There are also some reports on the high conductivity of RuO_2 that can be obtained just by including a small amount of spin-orbit coupling term into the calculation [14;22].

5 Conclusion

In summary, the investigation into altermagnetism within RuO_2 marks a notable progression in our comprehension of magnetic states, moving beyond the conventional categories of ferromagnetic and antiferromagnetic materials. This research has established the presence of altermagnetism in RuO_2, characterized by spin-splitting and an anti-aligned spin configuration, which is determined by symmetry rather than relying on spin-orbit coupling interactions. This discovery highlights the significance of such materials in bridging the realms of superconductivity and magnetic phenomena, indicating a profound connection that warrants further exploration. Additionally, we discussed the stability of on-site spins when electrons and holes are introduced into the system, offering insights into the interplay between magnetic states and occupied energy bands. We also proposed various avenues for continued research in this area, providing valuable tools for future studies.

The end

References

[1] A. J. Schofield. There and back again: from magnets to superconductors. *Physics*, 2:93, November 2009. Publisher: American Physical Society.

[2] P. Hohenberg and W. Kohn. Inhomogeneous Electron Gas. *Physical Review*, 136(3B):B864–B871, November 1964. Publisher: American Physical Society.

[3] John P. Perdew, J. A. Chevary, S. H. Vosko, Koblar A. Jackson, Mark R. Pederson, D. J. Singh, and Carlos Fiolhais. Atoms, molecules, solids, and surfaces: Applications of the generalized gradient approximation for exchange and correlation. *Physical Review B*, 46(11):6671–6687, September 1992. Publisher: American Physical Society.

[4] A. D. Becke. Density-functional exchange-energy approximation with correct asymptotic behavior. *Physical Review A*, 38(6):3098–3100, September 1988. Publisher: American Physical Society.

[5] David C. Langreth and M. J. Mehl. Beyond the local-density approximation in calculations of ground-state electronic properties. *Physical Review B*, 28(4):1809–1834, August 1983. Publisher: American Physical Society.

[6] Paolo Giannozzi, Stefano Baroni, Nicola Bonini, Matteo Calandra, Roberto Car, Carlo Cavazzoni, Davide Ceresoli, Guido L. Chiarotti, Matteo Cococcioni, Ismaila Dabo, Andrea Dal Corso, Stefano de Gironcoli, Stefano Fabris, Guido Fratesi, Ralph Gebauer, Uwe Gerstmann, Christos Gougoussis, Anton Kokalj, Michele Lazzeri, Layla Martin-Samos, Nicola Marzari, Francesco Mauri, Riccardo Mazzarello, Stefano Paolini, Alfredo Pasquarello, Lorenzo Paulatto, Carlo Sbraccia, Sandro Scandolo, Gabriele Sclauzero, Ari P. Seitsonen, Alexander Smogunov, Paolo Umari, and Renata M. Wentzcovitch. QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials. *Journal of Physics. Condensed Matter: An Institute of Physics Journal*, 21(39):395502, September 2009.

[7] P. Giannozzi, O. Andreussi, T. Brumme, O. Bunau, M. Buongiorno Nardelli, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, M. Cococcioni, N. Colonna, I. Carnimeo, A. Dal Corso, S. de Gironcoli, P. Delugas, R. A. DiStasio, A. Ferretti, A. Floris, G. Fratesi, G. Fugallo, R. Gebauer, U. Gerstmann, F. Giustino, T. Gorni, J. Jia, M. Kawamura, H.-Y. Ko, A. Kokalj, E. Küçükbenli, M. Lazzeri, M. Marsili, N. Marzari, F. Mauri, N. L. Nguyen, H.-V. Nguyen, A. Otero-de-la Roza, L. Paulatto, S. Poncé, D. Rocca, R. Sabatini, B. Santra, M. Schlipf, A. P. Seitsonen, A. Smogunov, I. Timrov, T. Thonhauser, P. Umari, N. Vast, X. Wu, and S. Baroni. Advanced capabilities for materials modelling with Quantum ESPRESSO. *Journal of Physics. Condensed Matter: An Institute of Physics Journal*, 29(46):465901, November 2017.

[8] Paolo Giannozzi, Oscar Baseggio, Pietro Bonfà, Davide Brunato, Roberto Car, Ivan Carnimeo, Carlo Cavazzoni, Stefano de Gironcoli, Pietro Delugas, Fabrizio Ferrari Ruffino, Andrea Ferretti, Nicola Marzari, Iurii Timrov, Andrea Urru, and Stefano Baroni. Quantum ESPRESSO toward the exascale. *The Journal of Chemical Physics*, 152(15):154105, April 2020.

[9] Nicola Marzari, Arash A. Mostofi, Jonathan R. Yates, Ivo Souza, and David Vanderbilt. Maximally localized Wannier functions: Theory and applications. *Reviews of Modern Physics*, 84(4):1419–1475, October 2012. Publisher: American Physical Society.

[10] Nicola Marzari and David Vanderbilt. Maximally localized generalized Wannier functions for composite energy bands. *Physical Review B*, 56(20):12847–12865, November 1997. Publisher: American Physical Society.

[11] E. I. Blount. Formalisms of Band Theory. In Frederick Seitz and David Turnbull, editors, *Solid State Physics*, volume 13, pages 305–373. Academic Press, January 1962.

[12] M. Born and V. Fock. Beweis des Adiabatensatzes. *Zeitschrift für Physik*, 51(3):165–180, March 1928.

[13] J. J. Sakurai and Jim Napolitano. Modern Quantum Mechanics, September 2020. ISBN: 9781108587280 Publisher: Cambridge University Press.

[14] Meng Wang, Katsuhiro Tanaka, Shiro Sakai, Ziqian Wang, Ke Deng, Yingjie Lyu, Cong Li, Di Tian, Shengchun Shen, Naoki Ogawa, Naoya Kanazawa, Pu Yu, Ryotaro Arita, and Fumitaka Kagawa. Emergent zero-field anomalous Hall effect in a reconstructed rutile antiferromagnetic metal. *Nature Communications*, 14(1):8240, December 2023. Publisher: Nature Publishing Group.

[15] Paul Erhart, Andreas Klein, Daniel Åberg, and Babak Sadigh. Efficacy of the DFT + U formalism for modeling hole polarons in perovskite oxides. *Physical Review B*, 90(3):035204, July 2014. Publisher: American Physical Society.

[16] J. Gaudet, A. C. Tavares, S. Trasatti, and D. Guay. Physicochemical Characterization of Mixed RuO2SnO2 Solid Solutions. *Chemistry of Materials*, 17(6):1570–1579, March 2005. Publisher: American Chemical Society.

[17] S.R. Butler and J.L. Gillson. Crystal growth, electrical resistivity and lattice parameters of RuO2 and IrO2. *Materials Research Bulletin*, 6(2):81–89, February 1971.

[18] Kelsey A. Stoerzinger, Oscar Diaz-Morales, Manuel Kolb, Reshma R. Rao, Rasmus Frydendal, Liang Qiao, Xiao Renshaw Wang, Niels Bendtsen Halck, Jan Rossmeisl, Heine A. Hansen, Tejs Vegge, Ifan E. L. Stephens, Marc T. M. Koper, and Yang Shao-Horn. Orientation-Dependent Oxygen Evolution on RuO_2 without Lattice Exchange. *ACS Energy Letters*, 2(4):876–881, April 2017.

[19] V Maurya, G Sharma, and K B Joshi. First-principles characterisation of structural and electronic properties of some RuO2 crystals. *Physica Scripta*, 96(5):055807, March 2021. Publisher: IOP Publishing.

[20] K Benyahia, Z Nabi, A Tadjer, and A Khalfi. Ab initio study of the structural and electronic properties of the complex structures of RuO2. *Physica B: Condensed Matter*, 339(1):1–10, November 2003.

[21] Wahyu Setyawan and Stefano Curtarolo. High-throughput electronic band structure calculations: challenges and tools. *Computational Materials Science*, 49(2):299–312, August 2010. arXiv:1004.2974 [cond-mat].

[22] Zexin Feng, Xiaorong Zhou, Libor Šmejkal, Lei Wu, Zengwei Zhu, Huixin Guo, Rafael González-Hernández, Xiaoning Wang, Han Yan, Peixin Qin, Xin Zhang, Haojiang Wu, Hongyu Chen, Ziang Meng, Li Liu, Zhengcai Xia, Jairo Sinova, Tomáš Jungwirth, and Zhiqi Liu. An anomalous Hall effect in altermagnetic ruthenium dioxide. *Nature Electronics*, 5(11):735–743, November 2022. Publisher: Nature Publishing Group.

A Projection of Density of States on Each Orbitals